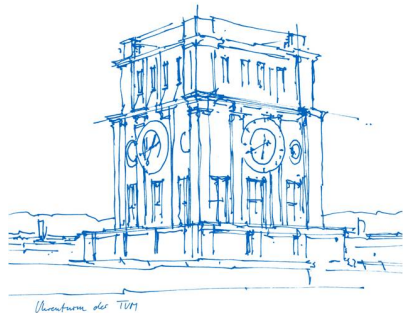


PoreAccess-CO₂: From Random Packing to an Interpretable Transport Closure

Structure-resolved diffusion and adsorption in
DEM-generated zeolite beds

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Presentation outline

Motivation and research question

Pore-network reference model

Reduction and diffusivity prediction

Results and robustness

Conclusions and next steps

Section 1 of 5

Motivation and research question

Why should inlet-connected pathways matter?

Why examine access from the inlet?

What porosity misses

- Porosity measures how much void space exists.
- It does not describe bottlenecks, path lengths or parallel connections.
- Similar void fractions can therefore hide different gas-supply pathways.

Why the inlet is the reference

- Gas enters from a reservoir below the bed.
- Adsorption removes gas as it diffuses into the packing.
- Access from the inlet may therefore help explain transient uptake differences.

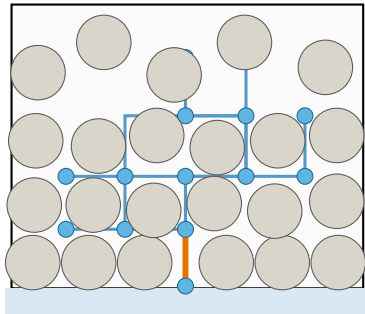
Application: CO₂ adsorption in zeolite 13X, used here as a controlled model problem.

Hypothesis: inlet-connected diffusive pathways contain information that porosity misses.

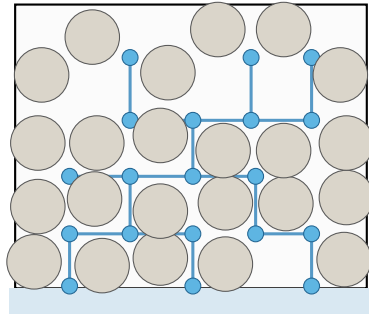
Same porosity can hide different inlet-connected pathways

Can a pore-accessibility measure predict bed-scale transient transport better than porosity alone?

Packing A: inlet bottleneck



Packing B: multiple inlet paths



- pore body
- diffusive throat
- zeolite bead
- inlet bottleneck

Same beads and similar porosity; different numbers and conductances of inlet-connected paths.

Section 2 of 5

Pore-network reference model

From particle geometry to the mean uptake curve

Twenty beds isolate variability caused by random arrangement

Quantity	Fixed value
beads per bed	1850
bead diameter	1.6 mm
tube diameter	16 mm
D_{tube}/d_p	10
temperature	298.15 K
inlet c_{CO_2}	6.05 mol m^{-3}
simulation time	5000 s

Controlled diffusion–adsorption problem

- same beads and tube; only the insertion seed changes
- dry, isothermal gas supplied from below
- diffusion and adsorption, with no imposed flow
- heat effects and competitive water adsorption excluded

$c_{\text{in}} = y_{\text{CO}_2} p / (RT) \approx 6.05 \text{ mol m}^{-3}$ at 15 vol.%, 1 bar and 298 K.

15% represents coal-combustion flue gas (typical CO_2 : 10–15%).

A dry, cooled reference gas fixes the driving concentration across all beds.

Source: U.S. DOE, Quadrennial Technology Review (2015), Sec. 4.E.

The ensemble tests arrangement effects under otherwise identical conditions.

Pore-network geometry and throat diffusion

Network construction

- four bead centres define a Delaunay tetrahedron; its valid circumcentre represents a pore node
- tetrahedra sharing a triangular face are joined by a throat
- topology comes from particle coordinates, not voxels
- 2^{22} Sobol points sample the solid and gas space.
- Gas-point counts allocate the calculated void volume to pores.

Diffusion through throat i - j

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$$

$D_m = 1.5 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$: molecular CO_2 diffusivity in the gas.

$L_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$ is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area

$$A_{ij} = \pi r_{ij}^2,$$

where r_{ij} is the clearance of the shared Delaunay face.

The circular throat area is a geometric modelling assumption; G_{ij} has units of $\text{m}^3 \text{ s}^{-1}$.

The pore network solves diffusion and adsorption together

Gas balance for each pore i

$$V_i \frac{dc_i}{dt} = \sum_{j \in \mathcal{N}(i)} G_{ij}(c_j - c_i) + G_{iR}(c_R - c_i) - S_i.$$

- V_i : gas volume, c_i : gas concentration
- G_{ij} : throat conductance [$\text{m}^3 \text{s}^{-1}$]
- G_{iR} : connection to reservoir at c_R , zero away from the inlet
- S_i : net gas transfer to adjacent beads [mol s^{-1}]

Uptake for each bead k

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

- q_k : current loading [mol kg^{-1}]
- k_{LDF} [s^{-1}]: linear-driving-force rate coefficient
- $1/k_{\text{LDF}}$: relaxation time at fixed gas concentration

One coupled system: one concentration per pore and one loading per bead.

Local equilibrium sets the adsorption target

Local equilibrium loading

$$p_k = c_k RT, \quad q_k^* = \frac{q_s b p_k}{1 + b p_k}.$$

- c_k : average gas concentration around bead k
- p_k : local CO₂ partial pressure [Pa]
- q_k^* : equilibrium target [mol kg⁻¹]

Fixed material parameters

- q_s : maximum CO₂ loading [mol kg⁻¹]
- b : Langmuir affinity [Pa⁻¹], so $b p_k$ is dimensionless
- k_{LDF} : kinetic coefficient [s⁻¹]

Actual loading evolves in time

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

Local gas concentrations set the target. Kinetics determine how quickly beads approach it.

The pore network delivers the reference uptake curve

Resolved transient outputs

- Gas concentration $c_i(t)$ in every pore
- Adsorbed loading $q_k(t)$ of every bead
- Different bead loadings reveal incomplete utilization

The detailed model retains the actual pore connections and bead associations.

Bed-average uptake

$$\bar{q}_{\text{PNM}}(t) = \frac{\sum_k m_k q_k(t)}{\sum_k m_k}.$$

For identical beads:

$$\bar{q}_{\text{PNM}}(t) = \frac{1}{N_b} \sum_{k=1}^{N_b} q_k(t).$$

This curve is the reference for fitting the reduced 1D model.

Next: can a homogeneous 1D bed reproduce this complete uptake curve?

Section 3 of 5

Reduction and diffusivity prediction

Fit a reduced model, then predict its coefficient

Reduce the detailed network to a homogeneous 1D bed

Detailed pore-network model

- Actual pores and throats from each packing
- One gas concentration per pore
- One loading per bead
- Reference mean uptake $\bar{q}_{\text{PNM}}(t)$

Homogeneous 1D model

- Replace the network by 100 equal axial slices
- Each slice spans the full tube cross-section
- One gas concentration and one mean solid loading per slice
- Represent unresolved pathways with a single diffusivity D

Fit D so the reduced model reproduces the network's mean uptake history.

The homogeneous 1D uptake model

Gas balance in each axial cell

$$\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t}.$$

Adsorbent loading in each axial cell

$$\frac{\partial q}{\partial t} = k_{\text{LDF}} \left[\frac{q_s b R T c}{1 + b R T c} - q \right].$$

- Divide the bed height into 100 equal slices, each spanning the tube cross-section.
- Solve gas concentration and mean solid loading together in each slice.
- Trial D changes c , which changes adsorption.

Average the calculated loading

$$\bar{q}_{1D}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz.$$

For 100 equal axial cells:

$$\bar{q}_{1D}(t; D) = \frac{1}{100} \sum_{\ell=1}^{100} q_{\ell}(t; D).$$

- Match bed height, porosity and inlet condition to the network.
- Keep isotherm, kinetics and initial state fixed. Only D changes.

The 1D grid is separate from the pore network.

Fit the effective diffusivity to the complete uptake curve

Per-bed optimization

1. Select a trial D within prescribed bounds.
2. Solve the coupled 1D equations from the same initial state.
3. Compare the full uptake curve with the PNM reference.
4. Use previous errors to select the next D and repeat (optimization).

Curve error and fitted coefficient

$$e_m(D) = \bar{q}_{1D}(t_m; D) - \bar{q}_{PNM}(t_m).$$

$$\text{NRMSE}(D) = \frac{\sqrt{\sum_m w_m e_m(D)^2}}{\bar{q}_{PNM}(t_f)},$$

$$\sum_m w_m = 1.$$

$$D_{\text{eff}} = \arg \min_D \text{NRMSE}(D).$$

The time weights account for unequal spacing of saved outputs.

A fitted D_{eff} is a reduction target for each bed, not yet a geometry-based prediction.

Can inlet accessibility predict the fitted diffusivity?

A steady geometric descriptor

- Use the same pore network with adsorption disabled.
- Fix concentrations at the inlet and an interior plane.
- Solve steady diffusion and sum the resulting molar flux $\dot{n}$.

$$G_{\text{access}} = \frac{\dot{n}}{\Delta c}, \quad D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

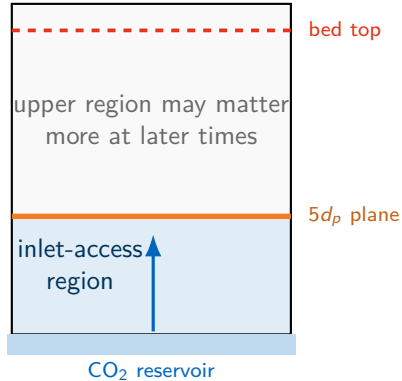
H : bed height; A_{tube} : tube cross-section.

Learn a transport closure

- Pair each D_{access} with its fitted D_{eff} .
- Fit $D_{\text{eff}} = f(D_{\text{access}})$ on training beds.
- Leave one bed out to test diffusivity prediction.
- For a new bed, apply the closure and run the 1D model.

Accessibility measures steady gas access. The closure links it to transient uptake.

Why use an interior plane?



A plane at 5d_p includes

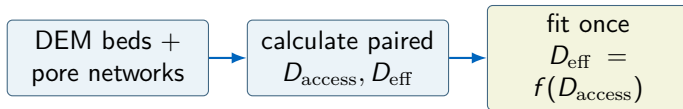
- the inlet openings
- second-layer and deeper bottlenecks
- parallel paths connecting inlet to the plane

The bed top answers a different question

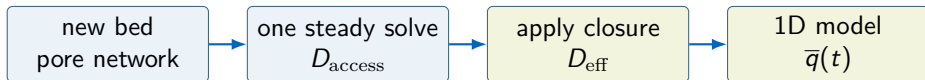
- measure conductance through the entire bed
- include upper bottlenecks that may matter more at later times

Development and later use are two different workflows

Develop closure



Predict for a new bed



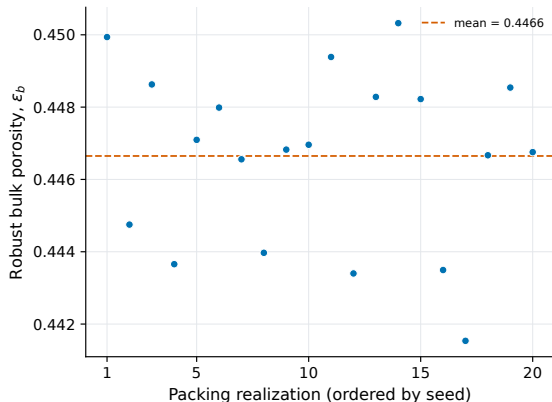
For a new bed, the transient pore-network adsorption model is no longer required.

Section 4 of 5

Results and robustness

Test the physical hypothesis and the model reduction

The twenty beds have nearly equal bulk porosity



Bulk-porosity CV: 0.55%.

Controlled ensemble

- same bead count and diameter
- same tube geometry
- only the insertion seed changes

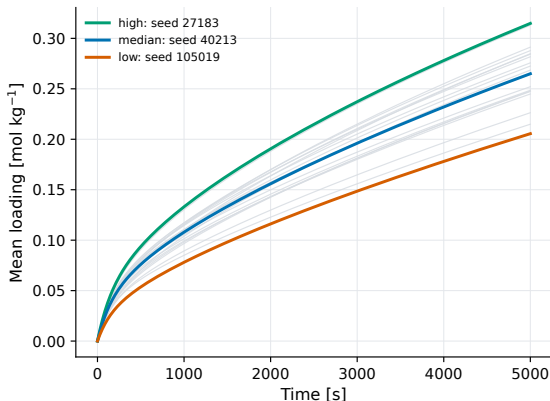
CV: coefficient of variation

$$CV_{\epsilon} = s_{\epsilon} / \bar{\epsilon}$$

s_{ϵ} : standard deviation

$\bar{\epsilon}$: mean porosity

Similar porosity still produces different uptake curves

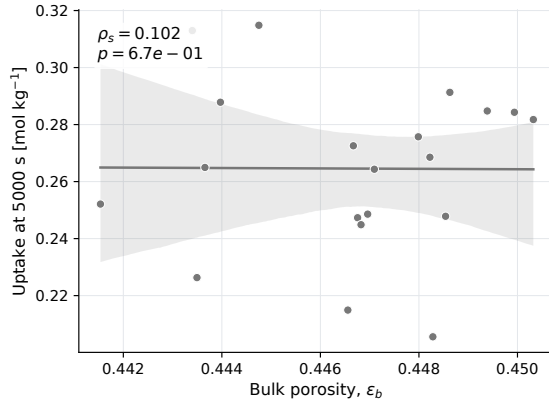


Network reference

- all twenty uptake curves
- highlighted beds span low, median and high final uptake
- identical adsorption parameters

At 5000 s: 0.206–0.315 mol kg⁻¹, with an uptake CV of 11.0%.

Bulk porosity alone does not explain final uptake

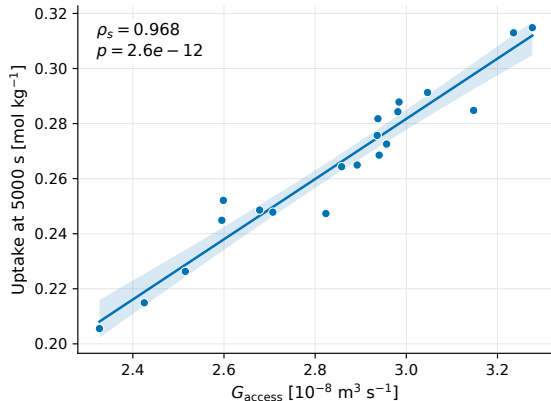


Porosity baseline

- each point represents one bed
- weak rank association
- wide uncertainty around the fitted relationship

Void volume alone does not describe access to the adsorbent.

Accessibility explains differences in final uptake



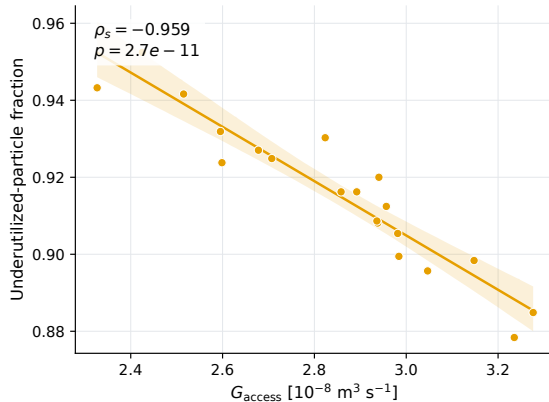
Steady network descriptor

- G_{access} measures inlet-to-interior conductance
- greater conductance accompanies greater uptake

Spearman $\rho_s = 0.968$.

Inlet-connected pathways explain variability that porosity misses.

Greater accessibility reduces particle underutilization



Higher G_{access} accompanies fewer underutilized particles.

Underutilized fraction

$$f_{\text{under}} = \frac{1}{N_p} \sum_{k=1}^{N_p} I_k.$$

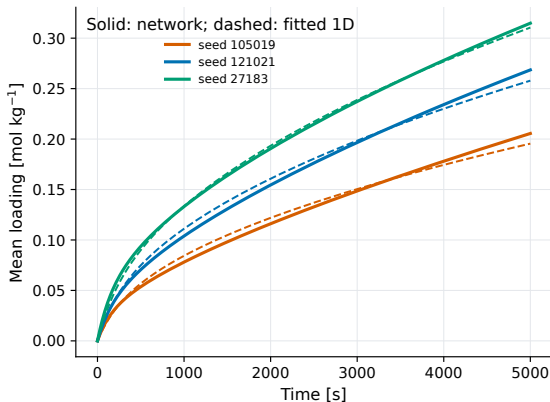
$I_k = 1$ when

$$q_k(t_f) < 0.5q^*(p_{\text{inlet}}).$$

$t_f = 5000$ s.

The indicator equals one for a particle below half the inlet-equilibrium loading.

One fitted D_{eff} reproduces each uptake curve



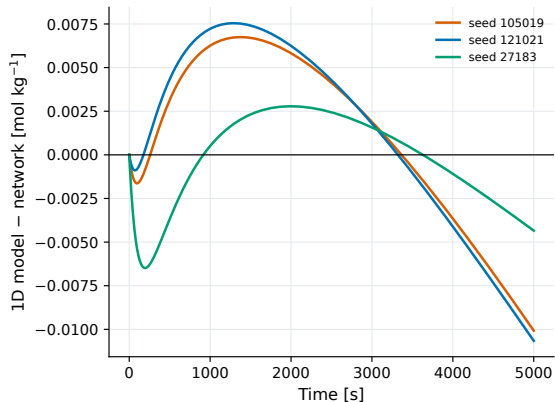
Per-bed optimization

1. solve the 1D model for a trial D
2. evaluate time-weighted NRMSE against the network curve
3. use previous errors to select the next D and repeat (optimization)

Solid: network.
Dashed: fitted 1D model.

Across twenty beds, the mean fitted-curve NRMSE is 1.58%.

Residuals show the remaining reduction error



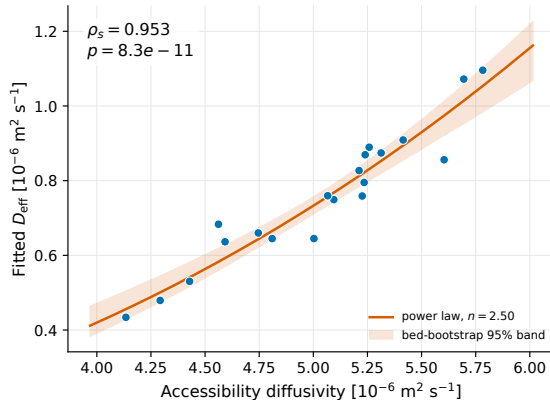
Curve residual

$$e(t) = \bar{q}_{1D}(t) - \bar{q}_{PNM}(t).$$

- positive: excess predicted uptake
- negative: insufficient predicted uptake

Time-weighted NRMSE across all beds: 0.84–2.46%.

A positive power law relates accessibility to diffusivity



Selected closure

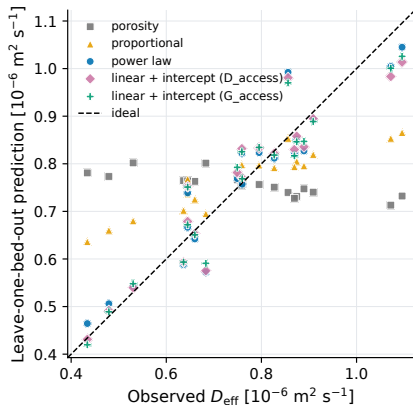
$$\frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n.$$

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

The shaded band shows bootstrap uncertainty in the fitted relationship.

The fitted exponent is $n \approx 2.50$ for this ensemble.

The closure predicts diffusivity for a withheld bed



Forms and held-out R^2

Predictor / form	R^2
$a_G G_{\text{access}} + b_G$	0.898
$CD_m(D_{\text{access}}/D_m)^n$	0.890
$a_D D_{\text{access}} + b_D$	0.881
αD_{access}	0.555
$a_0 + a_1 \varepsilon_b$	-0.234

Each expression predicts D_{eff} . Fit on nineteen beds and predict the omitted bed.

Power-law and linear-with-intercept scores are close; the power law is not the outright winner.

Why not predict uptake directly from G_{access} ?

Direct endpoint relation

$$G_{\text{access}} \longrightarrow \bar{q}(5000 \text{ s})$$

- strong correlation: $\rho_s = 0.968$
- ranks beds or predicts one endpoint
- valid at the selected time and conditions

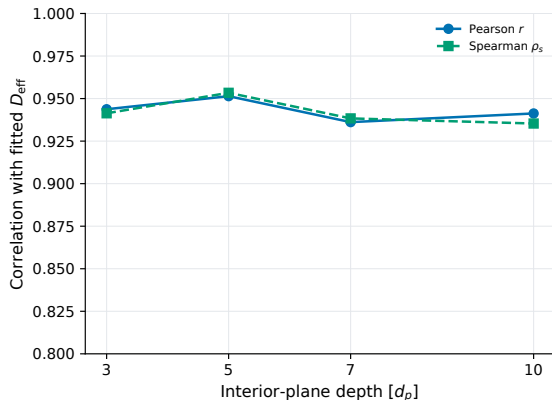
Transport closure

$$G_{\text{access}} \longrightarrow D_{\text{access}} \longrightarrow D_{\text{eff}} \longrightarrow \bar{q}(t)$$

- generates the complete uptake curve
- retains conservative 1D transport
- gives axial $c(z, t)$ and $q(z, t)$ profiles

Endpoint correlation summarizes one time. A transport closure supplies a complete uptake history.

Accessibility remains correlated across diagnostic depths



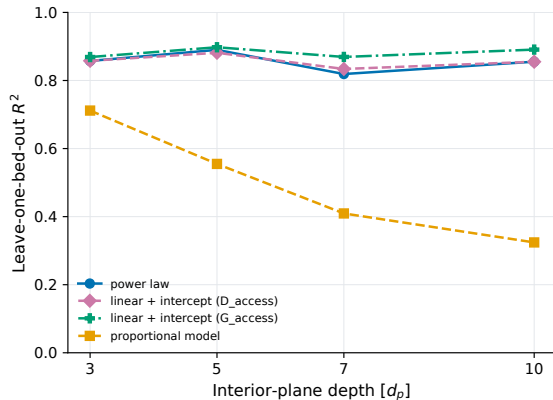
Depth sensitivity

- planes at 3, 5, 7 and $10d_p$
- Pearson: linear association
- Spearman: rank association

Depth is measured above each network's lowest pore.

Both correlations remain high across the tested depths.

Predictive performance across diagnostic depths



Four transport models

Power law

$$D_{\text{eff}} = CD_m(D_{\text{access}}/D_m)^n$$

Linear in D_{access} + intercept

$$D_{\text{eff}} = a_D D_{\text{access}} + b_D$$

Linear in G_{access} + intercept

$$D_{\text{eff}} = a_G G_{\text{access}} + b_G$$

Proportional

$$D_{\text{eff}} = \alpha D_{\text{access}}$$

Refit on nineteen beds at each depth; predict the omitted bed.

Power-law $R^2 = 0.819\text{--}0.890$ across depths; no consistent advantage over the linear models.

Section 5 of 5

Conclusions and next steps

What is established and what remains to validate

What the present evidence establishes

Supported

- porosity alone misses important packing variability
- inlet-connected topology controls transient utilization
- one D_{eff} reproduces each detailed curve accurately
- D_{access} provides an interpretable predictive closure

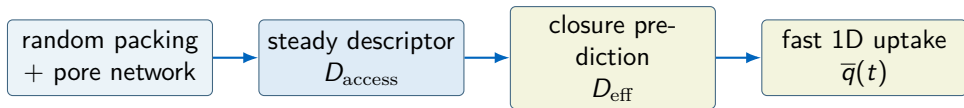
Not yet claimed

- industrial breakthrough performance
- effects of flow, heat or water competition
- transfer to other bead sizes or D_{tube}/d_p
- a universal power-law exponent

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

Pore accessibility retains what bulk porosity loses

Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



20 beds
controlled ensemble

1.58%
mean reduction error

$R^2 = 0.890$
withheld-bed closure